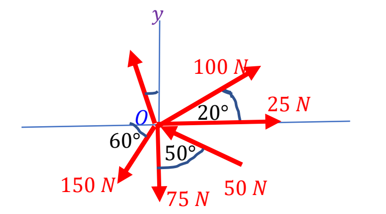
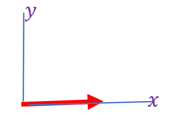
Concurrent forces in a plane

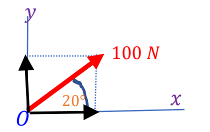
Consider a system of concurrent forces acting at point O. Let us find out the resultant of the forces. Let us take one force at a time and resolve it in to two components along the x-axis and y-axis. As per the sign convention, the forces acting along the positive co-ordinate directions are considered positive.





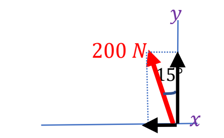
$F_{1x} = +25\,N$

$F_{1y} = 0$



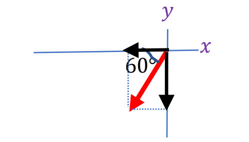
$F_{2x} = +100\cos 20 = +94N$

$F_{2y} = +100\sin 20 = +34N$



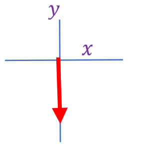
$F_{3x} = -200\sin 15 = -51.7N$

$F_{3y} = +200\cos 15 = +193.2N$



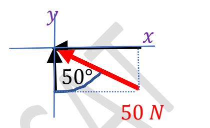
$F_{4x} = -150\cos 60 = -75N$

$F_{4y} = -150\sin 60 = -130N$



$F_{5x} = 0$

$F_{5y} = -75\,N$



$F_{6x} = -50\sin 50 = -38.3N$

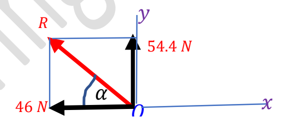
$F_{6y} = +50\cos 50 = +32.2N$

Now, all the forces are split in to two components; one along the -axis and the other along y-axis.

$\Sigma F_x = 25 + 94 - 51.7 - 75 + 0 - 38.3 = -46\,N$

$\Sigma F_y = 0 + 34 + 193.2 - 130 - 75 + 32.2 = 54.4N$

All the forces acting along the x-axis from a point O are combined to find the net force ΣF_x. Similarly, all the forces acting along the y-axis from a point O are combined to find the net force ΣF_y.

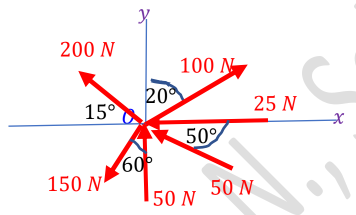


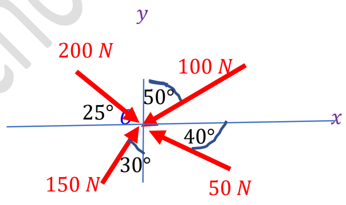
The resultant of these two perpendicular forces, $R = \sqrt{\Sigma {F_x}^2 + \Sigma {F_y}^2} = \sqrt{(-46)^2 + (54.4)^2} = 71.4\,N$

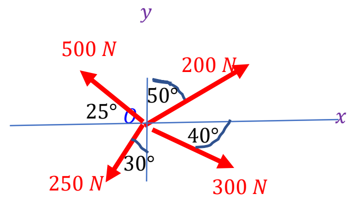
$$\tan\alpha = \frac{\Sigma F_y}{\Sigma F_x} = \frac{54.4}{46} = 1.18 \ \ggg\ \alpha = 49.8°$$

Exercise 1

Find the resultant of the concurrent force systems shown below.

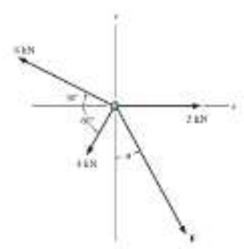






Exercise 2

Find the magnitude and direction of F so that the resultant of the concurrent force system is zero.



Reference

1. Engineering Mechanics: Timoshenko and Young, McGraw Hill
2. Mechanics for Engineers: Beer & Johnston, McGraw Hill
3. Engineering Mechanics: Merriam & Kraig, Wiley
4. Engineering Mechanics: Hibbeler, Pearson